# Gaussian Distribution

$N(\mu, \sigma^2)$

Calculate PDF of Gaussian distrib. using the following data $x = 2$, $\mu = 5$ and $\sigma = 3$

Parameters: $\mu \in R$ = mean

$\sigma^2 \in R_{>0}$ = variance

Support: $x \in R$

PDF: $\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

CDF: $\phi\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right]$

Error function

Quantile: $\mu + \sigma\sqrt{2}\, \text{erf}^{-1}(2p-1)$

Mean, Median, Mode: $\mu$

Variance: $\sigma^2$

$$e^{-\frac{(x-\mu)^2}{2\sigma^2}} \Rightarrow e^{-\frac{(2-5)^2}{18}}$$

$$\Rightarrow \frac{1}{(3\times 2.51)} \times 0.6065$$

$$\Rightarrow 0.1328 \times 0.6065$$

$$\Rightarrow 0.0805$$

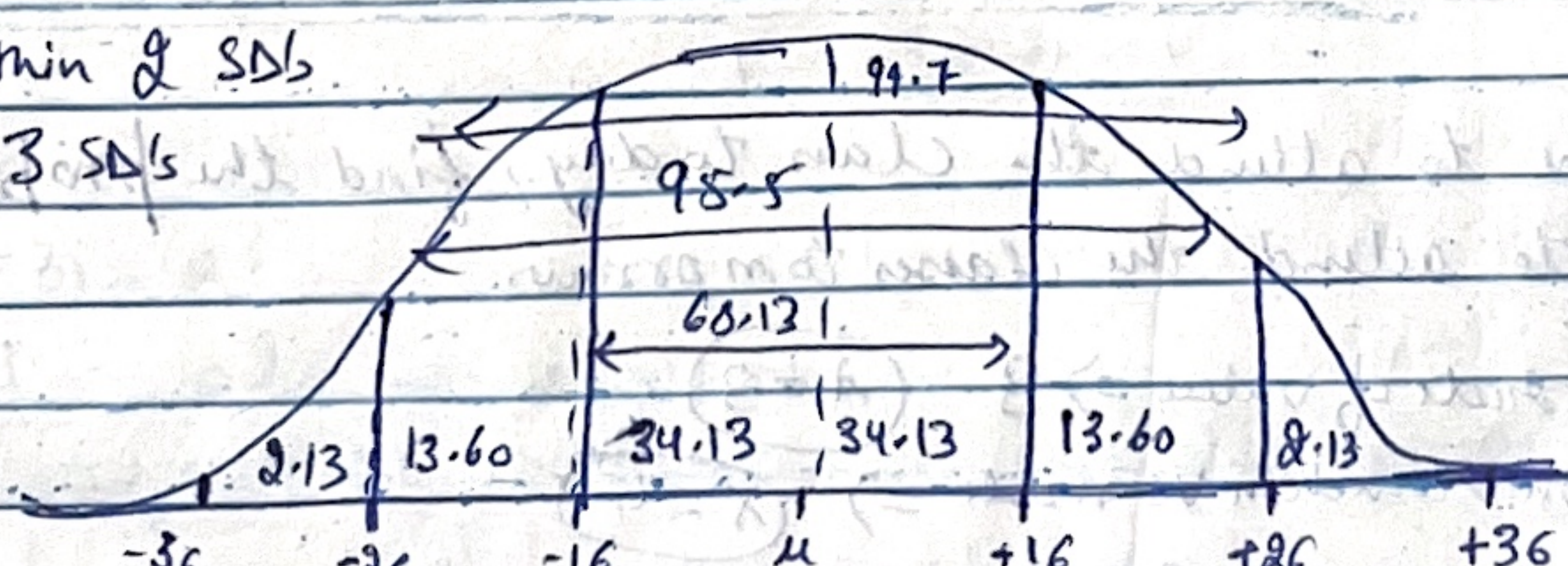
## Emperical Rule of Normal Distribution

Also known as 68 95 99 rule.

68% $\Rightarrow$ observed data points will lie inside one SD of the mean

95% $\Rightarrow$ fall within 2 SDs

99.7% $\Rightarrow$ within 3 SDs




68.13 → 1st SD of 1st mean

95.5 → 2nd SD of 2nd mean

99.7 $\Rightarrow$ 3rd SD of 3rd mean

## Bernoulli Distribution

Events that have one trial and two possible outcomes.

Mean $= p$

PMF $= \begin{cases} q = 1-p & \text{if } k=0 \\ p & \text{if } k=1 \end{cases}$

CDF $= \begin{cases} 0 & \text{if } k<0 \\ 1-p & \text{if } 0 \le k < 1 \\ 1 & \text{if } k \ge 1 \end{cases}$

Median $= \begin{cases} 0 & \text{if } p < 1/2 \\ [0,1] & \text{if } p = 1/2 \\ 1 & \text{if } p > 1/2 \end{cases}$

Mode $= \begin{cases} 0 & \text{if } p < 1/2 \\ 0,1 & \text{if } p = 1/2 \\ 1 & \text{if } p > 1/2 \end{cases}$

Variance $= p(1-p) = pq$

Without replacement

After 1 white drawn:- 9 white + 10 black left (19)

Now Black removes 10 white + 9 black

$P(W_2|W_1) = \frac{9}{19} \approx 0.4737$

$P(W_2|B_1) = \frac{10}{19} \approx 0.5263$

Probability of success is not constant

8 balls are drawn from bag containing 10 white and 10 black balls. Predict whether the trials are Bernoulli trials if ball drawn is replaced and not replaced.

a) When ball is drawn with replacement:

probability of success :- $p = \frac{10}{20} = 1/2$

b) When ball is drawn without replacement:

P(success) for trial 1 $= 1/2$

for trial 2 $= 9/19$

Not Bernoulli trial

With Replacement

$P(W_1) = 10/20 = 0.5$

$P(W_1 \text{ and } W_2) = 0.5 \times 0.5 = 0.25$